

CHAPTER 4 — CHECKING A PALINDROME NUMBER

SLIDE 1 — WHAT IS A PALINDROME NUMBER?

MEET PALINDROME NUMBERS!

Sometimes, a number looks **the same when we read it forward and backward**.

For example:

121

Forward:

1 → 2 → 1

Backward:

1 → 2 → 1

So, 121 is a **Palindrome Number**.

Simple Definition

A palindrome number is a number that remains the same when its digits are reversed.

LOOK AT THIS!

121

1 2 1

↓ ↓ ↓

1 2 1

Same from both sides!

So:

121 → Palindrome ✓

But:

123 → 321

So:

123 → Not Palindrome ✗

EASY ANALOGY

Think of a palindrome like a **mirror** 🪞

1 2 1

↓ 🪞 ↓

1 2 1

The number looks the same!

★ REMEMBER!

Same after reversing → Palindrome ✓

Different after reversing → Not Palindrome ✗

SLIDE 2 — REVERSING THE NUMBER

HOW DO WE CHECK A PALINDROME?

First, we need to **reverse the number**.

We already know:

% 10 → Get the last digit 🔍

/ 10 → Remove the last digit ÷

EXAMPLE

Let's take:

121

STEP 1

None

$$121 \% 10 = 1$$

$$121 / 10 = 12$$

👉 Digit = 1

STEP 2

None

$$12 \% 10 = 2$$

$$12 / 10 = 1$$

👉 Digit = 2

STEP 3

None

$$1 \% 10 = 1$$

$$1 / 10 = 0$$

👉 Digit = 1

🔍 SO WE GET:

1 → 2 → 1

Build the reverse:

121

 RESULT

None

Original → 121

Reverse → 121

They are the same!

👉 Palindrome ✅

★ REMEMBER!

First reverse the number ↺

Then compare it with the original ⚖️

🟡 SLIDE 3 — ⚖️ COMPARING ORIGINAL AND REVERSE

🤔 WHY DO WE SAVE THE ORIGINAL?

While reversing, the original number keeps changing.

For example:

None

121 → 12 → 1 → 0

So, we need to **save the original number** before changing it.

📝 SAVE THE ORIGINAL

None

```
original = n;
```

Now we have:

None

`original` → 121

`n` → 121

After reversing:

None

`original` → 121

`rev` → 121

NOW COMPARE

None

`if (original == rev)`

If they are equal:

👉 **Palindrome**

Otherwise:

👉 **Not Palindrome**

EASY ANALOGY

Think of it like checking two photos 📷

📷 Original → 121

📷 Reverse → 121

Compare them:

Same → 

Different → 

MEMORY TRICK

SAVE → REVERSE → COMPARE

★ TAKEAWAY

Always save the original number before reversing it.

SLIDE 4 — CHECKING PALINDROME IN C

🤔 CAN WE DO EVERYTHING USING A LOOP?

Yes! 🎯

We can use the same digit-accessing technique.

BASIC CODE

```
None
int n, original, digit, rev = 0;

scanf("%d", &n);

original = n;

while (n != 0)
{
    digit = n % 10;
    rev = rev * 10 + digit;
    n = n / 10;
}

if (original == rev)
{
    printf("Palindrome");
}
else
{
    printf("Not Palindrome");
}
```

```
}
```

HOW DOES IT WORK?

For:

n = 121

None

```
original = 121
```

Then:

None

```
121 → digit = 1 → rev = 1  
12 → digit = 2 → rev = 12  
1 → digit = 1 → rev = 121
```

Finally:

None

```
original = 121  
rev      = 121
```

So:

Palindrome 

BIG MEMORY TRICK

% 10 → GET 

/ 10 → REMOVE 

rev * 10 + digit → BUILD 

original == rev → CHECK 

 **TAKEAWAY**

Save the original → Reverse the number → Compare both numbers.

SLIDE 5 — PALINDROME RECAP

WHAT DID WE LEARN?

1. SAVE THE ORIGINAL

None

```
original = n;
```

Keep a copy before changing `n`.

2. GET THE LAST DIGIT

None

```
digit = n % 10;
```

`% 10` → GET 

3. REMOVE THE LAST DIGIT

None

```
n = n / 10;
```

`/ 10` → REMOVE 

4. BUILD THE REVERSE

None

```
rev = rev * 10 + digit;
```

Build the reversed number 

5. COMPARE

None

```
if (original == rev)
```

Same → **Palindrome** 

Different → **Not Palindrome** 

ONE BIG MEMORY TRICK!

None

SAVE

↓

REVERSE

↓

COMPARE

↓

PALINDROME? 

Question

1 What is a Palindrome Number?

Which of the following is a palindrome?

- A) 123
- B) 121
- C) 1234
- D) 456



Hint: Reverse the number and check if it stays the same.

 Glossary:

Palindrome → A number that remains the same when reversed.

2 Why do we save the original number?

None

```
int n = 121;  
int original;  
original = n;
```

Why do we use `original = n;`?

- A) To reverse the number
- B) To save a copy before changing `n`
- C) To remove the last digit
- D) To count the digits

 Hint: `n` changes while we reverse it.

 Glossary:

Original → The number before any changes are made.

3 What should be true for a number to be a Palindrome?

- A) Original > Reverse
- B) Original < Reverse
- C) Original == Reverse
- D) Original != Reverse

 Hint: A palindrome looks the same in both directions.

 Glossary:

`==` → Checks whether two values are equal.

4 Which statement correctly describes a Palindrome Number?

- A) It has exactly two factors.
- B) Its digits add up to the number.
- C) It remains the same when reversed.
- D) It always has an even number of digits.

 Hint: Think about the mirror analogy .

 Glossary:

Palindrome Number → A number that remains unchanged when its digits are reversed.

5 Find the Mistake

A student writes:

None

```
if (original != rev)
{
    printf("Palindrome");
}
```

What is wrong?

- A) `original` should be `digit`
- B) `!=` should be `==`
- C) `rev` should be `n`
- D) Nothing is wrong

💡 Hint: A palindrome has the same original and reverse.

📖 Glossary:

$\neq$ → Means “not equal to”.